DATERA EDU: Project Management and Risks

1. **Team and working procedures**

Three students and a supervisor. Team meets the supervisor every week.

- Tasks are kept in **YouTrack** - https://yakarta.youtrack.cloud/agiles/177-28/current
- **FIGMA** - https://www.figma.com/design/oVOPB0JhPKVFBVWMuBbSlG/DATERA-EDU---UI?node-id=28-6&t=PEkTWHH2R80kr88c-1

- Code is kept in **GitHub** - https://github.com/ETairkhan/diplomka

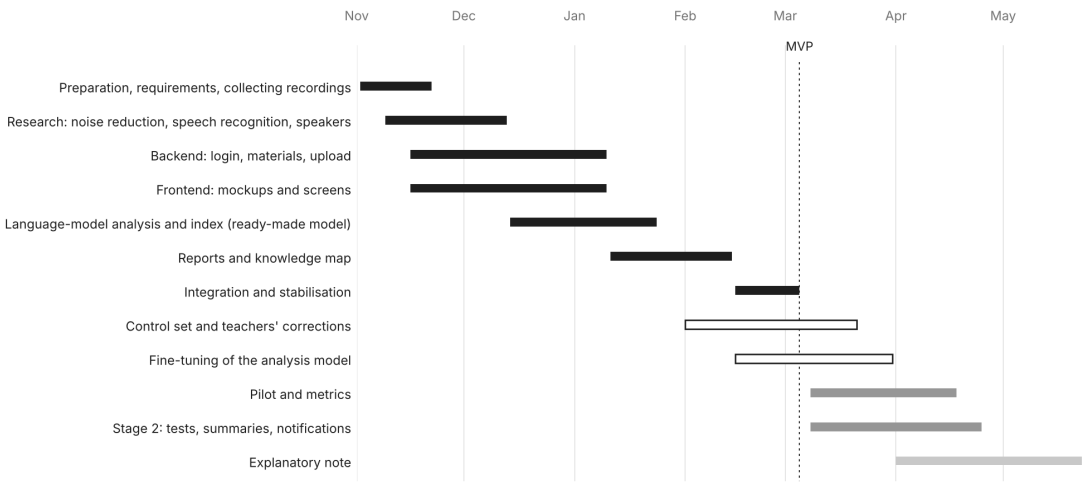
Who	Role	Owns
David Khamitskiy	Project manager, ML/AI	Requirements and priorities; speech pipeline; language-model analysis and fine-tuning; collecting teachers' ratings
Tairkhan Yermek	Backend developer	Go API, database, task queue, file storage, login, connection to the ML service
Yermakhan Khamitov	Frontend developer	Web interface for three roles, reports, knowledge map, audio player

Supervisor: Khaimuldin Nursultan Gazizovich, Astana IT University.

- Once a month the tasks in YouTrack are reviewed and the distribution of work is changed if needed.
- Code is in Git. Linters and tests run on every change.
- After each weekly meeting with the supervisor, the topics discussed and the decisions made are recorded.

2. **Plan**

The MVP is ready in early March 2027. Fine-tuning is planned for late winter to early spring.



- Nov 2026: requirements approved, development starts, first test recordings.
- Nov 2026 - Feb 2027: backend, interface, analysis, reports, knowledge map.
- Late winter - early spring 2027: fine-tuning of the analysis model.
- Early Mar 2027: MVP on ready-made models.
- Mar - Apr 2027: pilot with 2-3 teachers; second stage (tests, summaries, notifications).
- Apr - May 2027: explanatory note.

3. Main risks

The biggest risks are not about code: a GPU server and enough real lessons with consent are needed. The full list with fallbacks is in section 9.1 of the technical specification.

Risk	What could happen	Mitigation	Owner	Level
No GPU server	The whole pipeline cannot be run on real lessons	Agree on access (university server or rented GPU in Kazakhstan) in the first weeks of November. Meanwhile, research only on open data in Colab or Kaggle	David	High
Too few real lessons with consent	Fewer than the 20 recordings required; children's recordings need a parent's consent	Start with the teachers already interviewed. Have the consent text ready before the first upload	David	High
Speech recognition is worse on real lessons	More errors, especially on mixed English-Russian speech	Measure the error rate on the first recordings in November, compare two Whisper variants, mark low-accuracy lessons	David	Medium
Fine-tuning does not	Fewer than 300 examples by spring,	Trial run on open datasets first. Collect teachers' corrections from	David	Medium

Risk	What could happen	Mitigation	Owner	Level
give enough	or the tuned model is worse than the cloud one	the first lesson. If the result is weak, keep the cloud model and describe the comparison honestly		
Teachers do not trust the ratings	One teacher said she would stop using a system that makes frequent mistakes	The teacher can confirm or change any rating. Each rating shows its place in the recording. Target: at least 70 % of ratings stay unchanged	David	Medium
Project manager is overloaded	David covers the ML pipeline, the language model, fine-tuning and ratings	Monthly review of tasks. The backend developer takes the connection of the ML service to the queue	All	Medium
MVP is late	The core cannot be shown at the end of winter	The core is protected, the second stage is cut first (see Fallback plan)	All	Medium

4. Fallback plan

The MVP is protected: upload of materials and a lesson, analysis, three reports and the knowledge map. The second stage is cut first, in this order:

1. Notifications and weekly summaries.
2. Index graph and risk level.
3. Import of the question bank.
4. The tests themselves.

If the MVP itself is late, the screens are simplified, not the processing pipeline. If too few examples are collected for fine-tuning, the model is trained on the available examples and this is stated in the thesis.

5. Dependencies

The project depends on the following external conditions. Each has a fallback, so a missing item slows the work but does not stop it.

Dependency	Provided by	Needed by	If not available
GPU server for fine-tuning	University or a rented server in Kazakhstan	November 2026	Experiments on open data in Colab or Kaggle; real lessons are processed only on a server in Kazakhstan
Recordings of real lessons with consent (at least 20)	Teachers and parents	First tests	Start with the teachers already interviewed
2-3 pilot teachers	Team and supervisor	March 2027	Fewer teachers; results are shown as a case study
Cloud LLM API	External provider	MVP	Switch to another provider through the LLM adapter
Legal check of the consent text and privacy policy	University or a lawyer	Before the pilot	The pilot does not start until the check is done
Supervisor approval of the proposed targets	Supervisor	Before the pilot	Targets stay marked as proposed

6. Open items

- GPU server: who gives access and from when.
- Legal check of the consent text and privacy policy before the pilot.
- Pilot: confirm 2-3 teachers who will upload materials and rate 5-10 lessons.
- Start the log of supervisor meetings.